EXPERIMENT 1

A. SHELL SCRIPT – ARITHMETIC OPERATION USING expr COMMAND

AIM: To perform basic arithmetic operations using the expr command in a shell script.

SOFTWARE REQUIRED: Oracle VirtualBox, Fedora Linux, Terminal, Bash shell.

PROCEDURE:

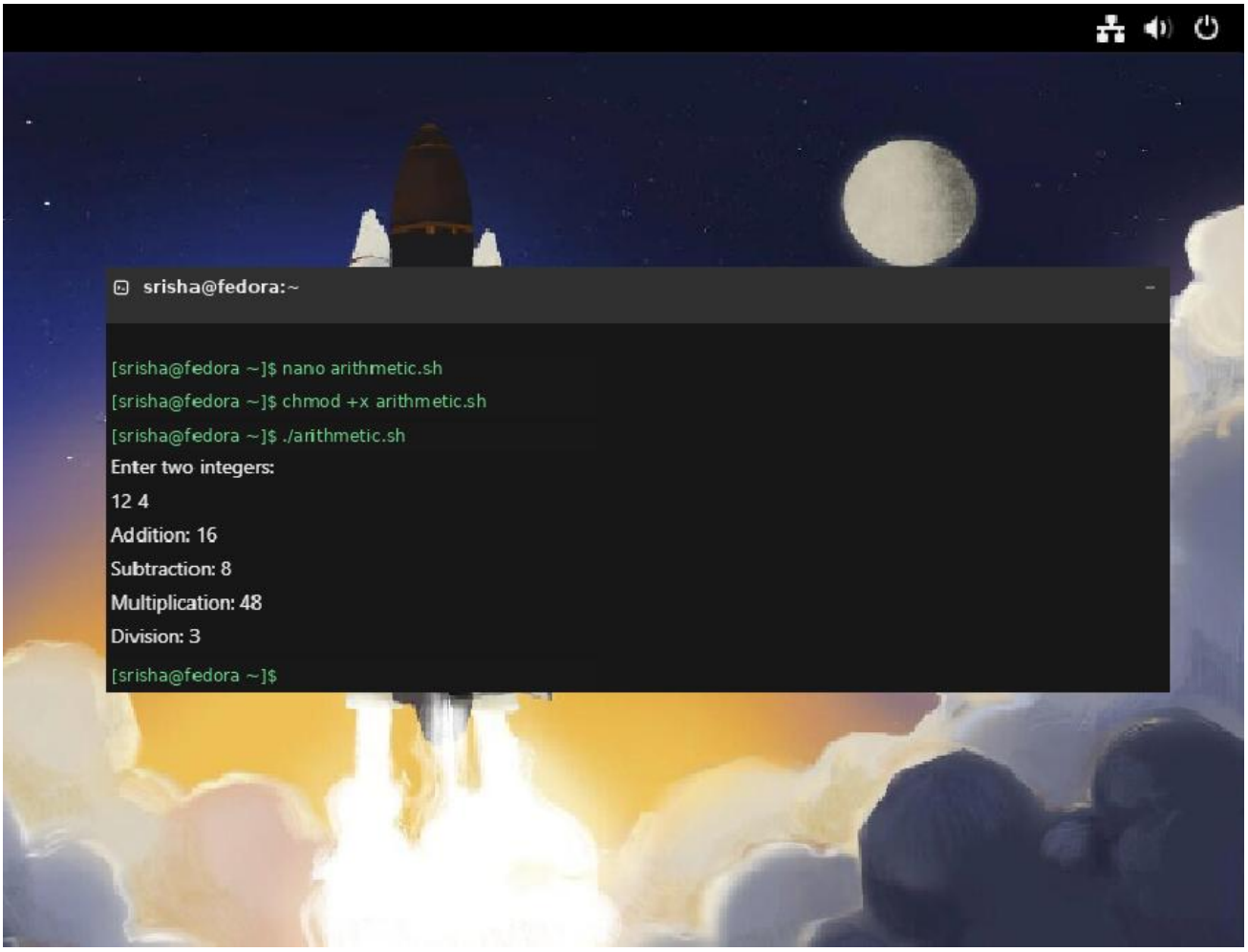
1. Open the Fedora terminal in Oracle VirtualBox.
2. Create a script using nano arithmetic.sh.
3. Enter the script and save the file.
4. Make it executable using chmod +x arithmetic.sh.
5. Run it using ./arithmetic.sh and enter two integers.

PROGRAM:

```
echo "Enter two integers:"
read a b
echo "Addition: $(expr "$a" + "$b")"
echo "Subtraction: $(expr "$a" - "$b")"
echo "Multiplication: $(expr "$a" \* "$b")"
if [ "$b" -ne 0 ]; then
    echo "Division: $(expr "$a" / "$b")"
else
    echo "Division is not possible (division by zero)."
Fi
```

OUTPUT:

```
Enter two integers:
12 4
Addition: 16
Subtraction: 8
Multiplication: 48
Division: 3
```



RESULT: Thus, the shell script performed arithmetic operations using the expr command.

EXPERIMENT 1

B.SHELL SCRIPT – CHECK LEAP YEAR USING if-else

AIM: To determine whether a given year is a leap year using if-else statements in a shell script.

SOFTWARE REQUIRED: Oracle VirtualBox, Fedora Linux, Terminal, Bash shell.

PROCEDURE:

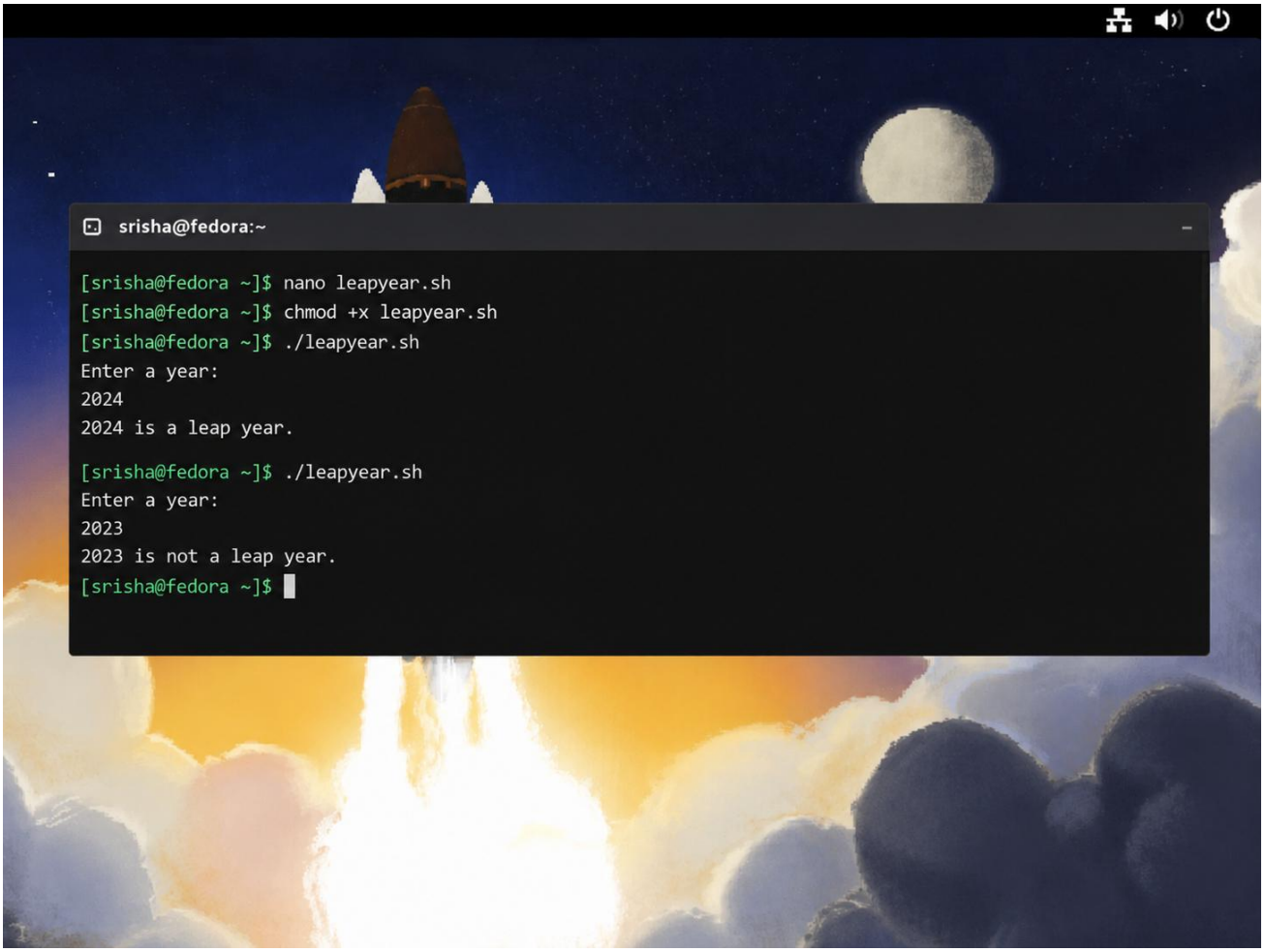
1. Open the Fedora terminal and create leapyear.sh.
2. Type the program, save it, and make it executable with chmod +x leapyear.sh.
3. Run ./leapyear.sh and enter a year.

PROGRAM:

```
echo "Enter a year:"
read year
if [ $((year % 400)) -eq 0 ] || { [ $((year % 4)) -eq 0 ]
&& [ $((year % 100)) -ne 0 ]; }; then
    echo "$year is a leap year."
else
    echo "$year is not a leap year."
fi
```

OUTPUT:

```
Enter a year:
2024
2024 is a leap year.
```



RESULT: Thus, the script correctly checked whether the entered year was a leap year.